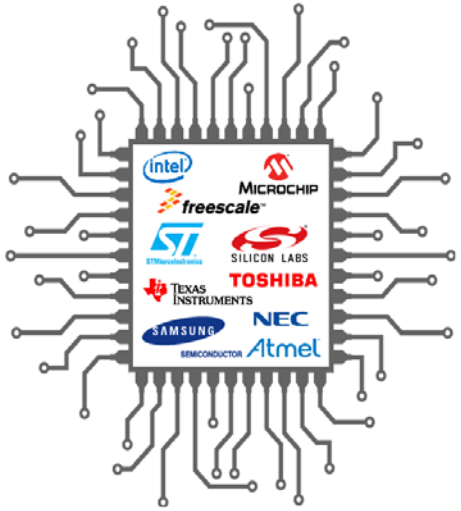
Introduction to Embedded Systems



Thareendhra Keerthi Wijayasiriwardhane

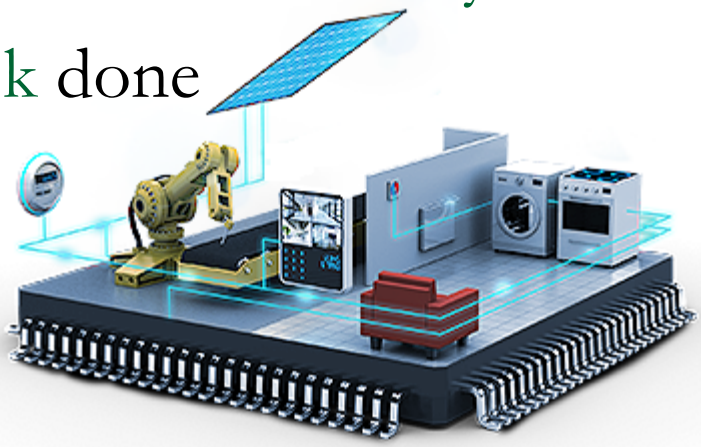
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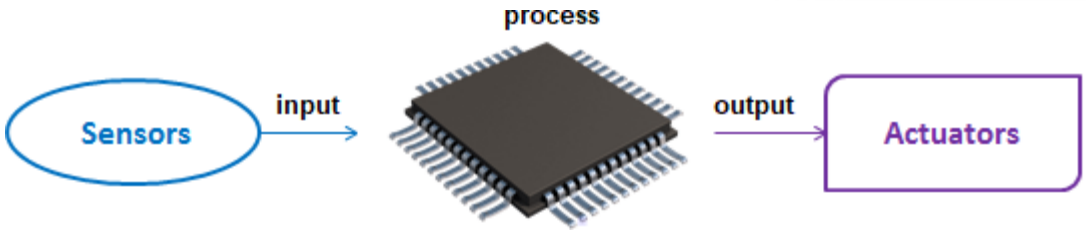
Learning Outcomes

- On completion of this session, you should be able to
  - Describe what an embedded system is
  - Explain how embedded systems can sense and change the state of physical world
  - Discuss about the constraints on embedded systems
  - Discuss about the evolution of embedded systems
  - Explain the growth in the use of embedded systems
  - Identify the components of an embedded system
  - Compare different implementations of embedded systems

Embedded System

- Computer system embedded into some other system
- Used to get some dedicated task done
- Preforms Physical Computing
  - Sensors
  - Processor
  - Actuators





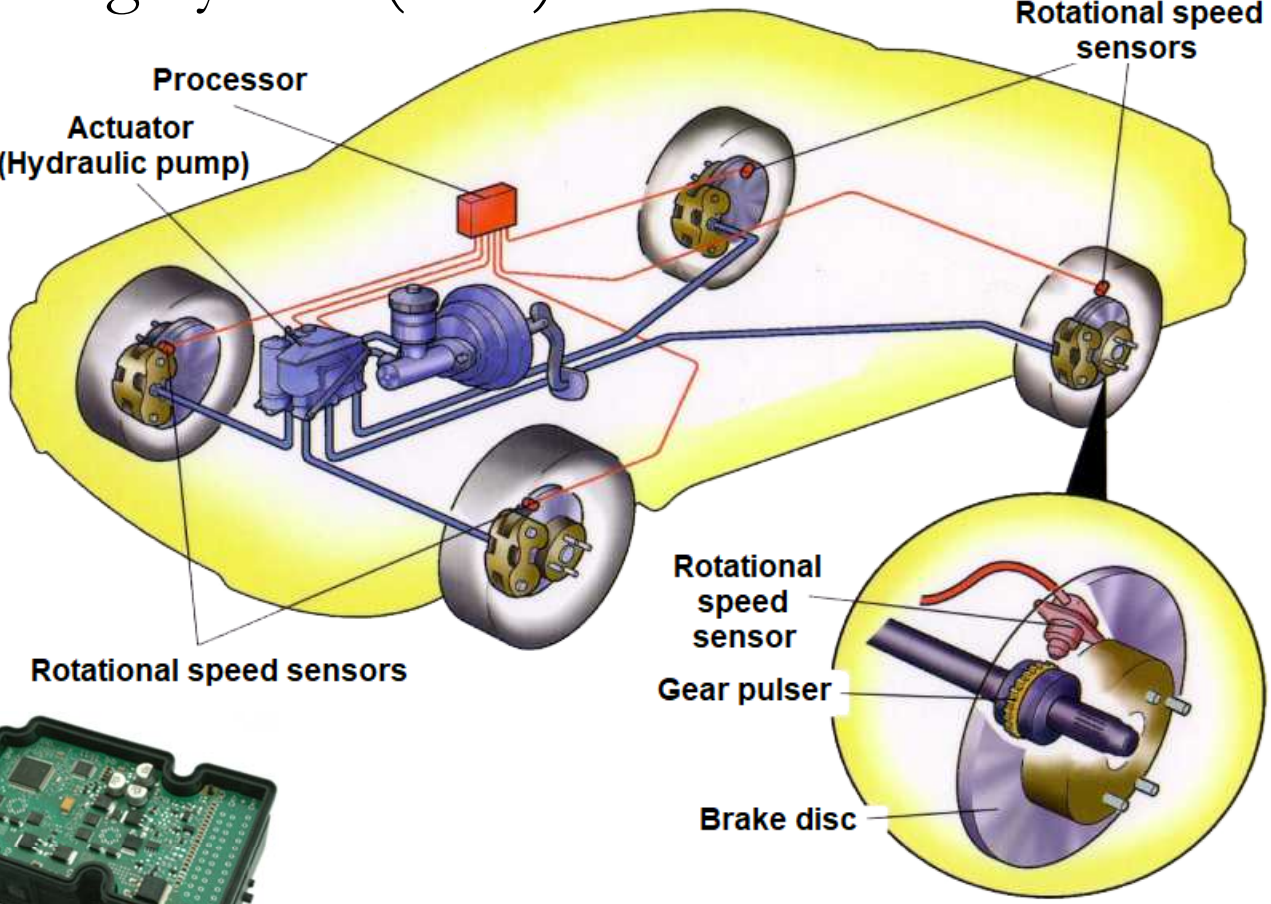
- Usually operates under strict constraints
  - What are they?

Example

- Anti-lock Braking System (ABS)
  - What is the main task?
  - How is it achieved?



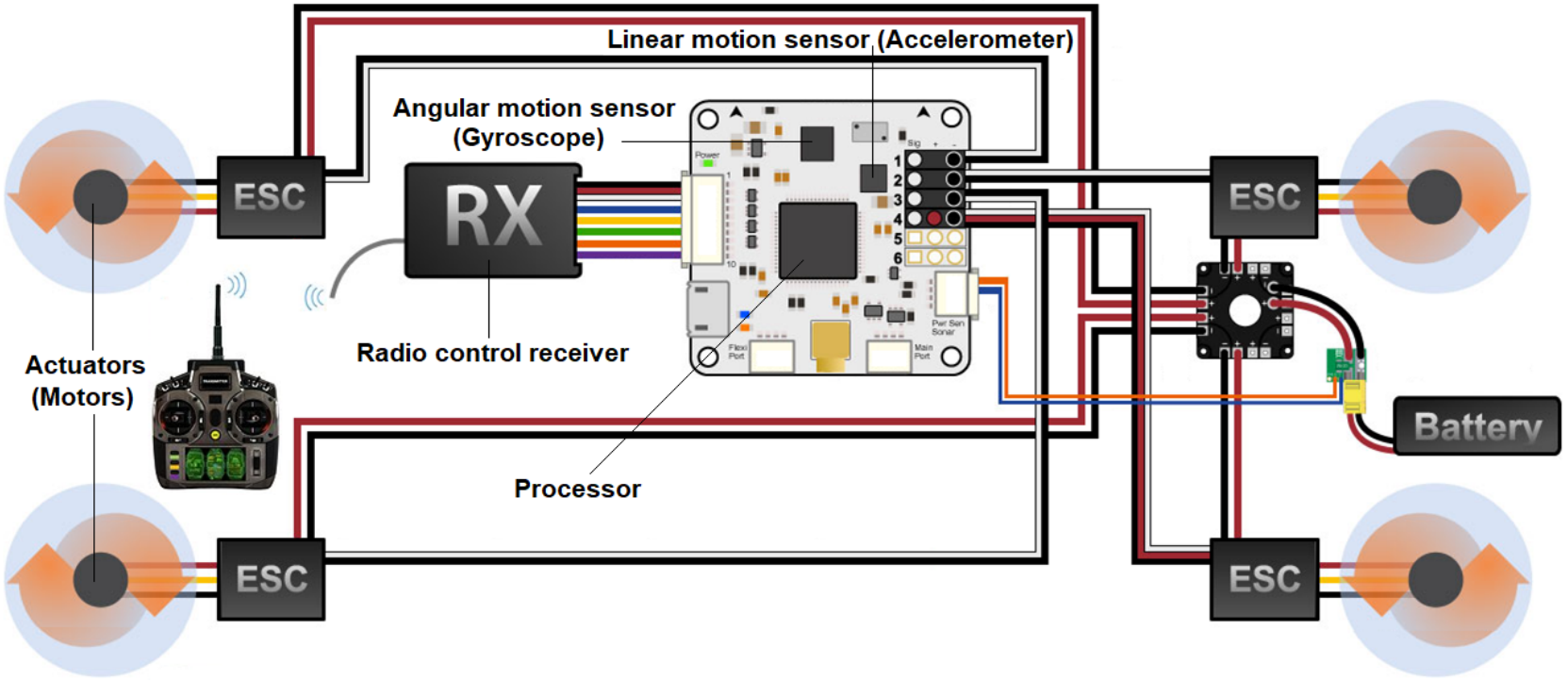




Example

- Flight Controller System
  - What is the main task?
  - How is it achieved?





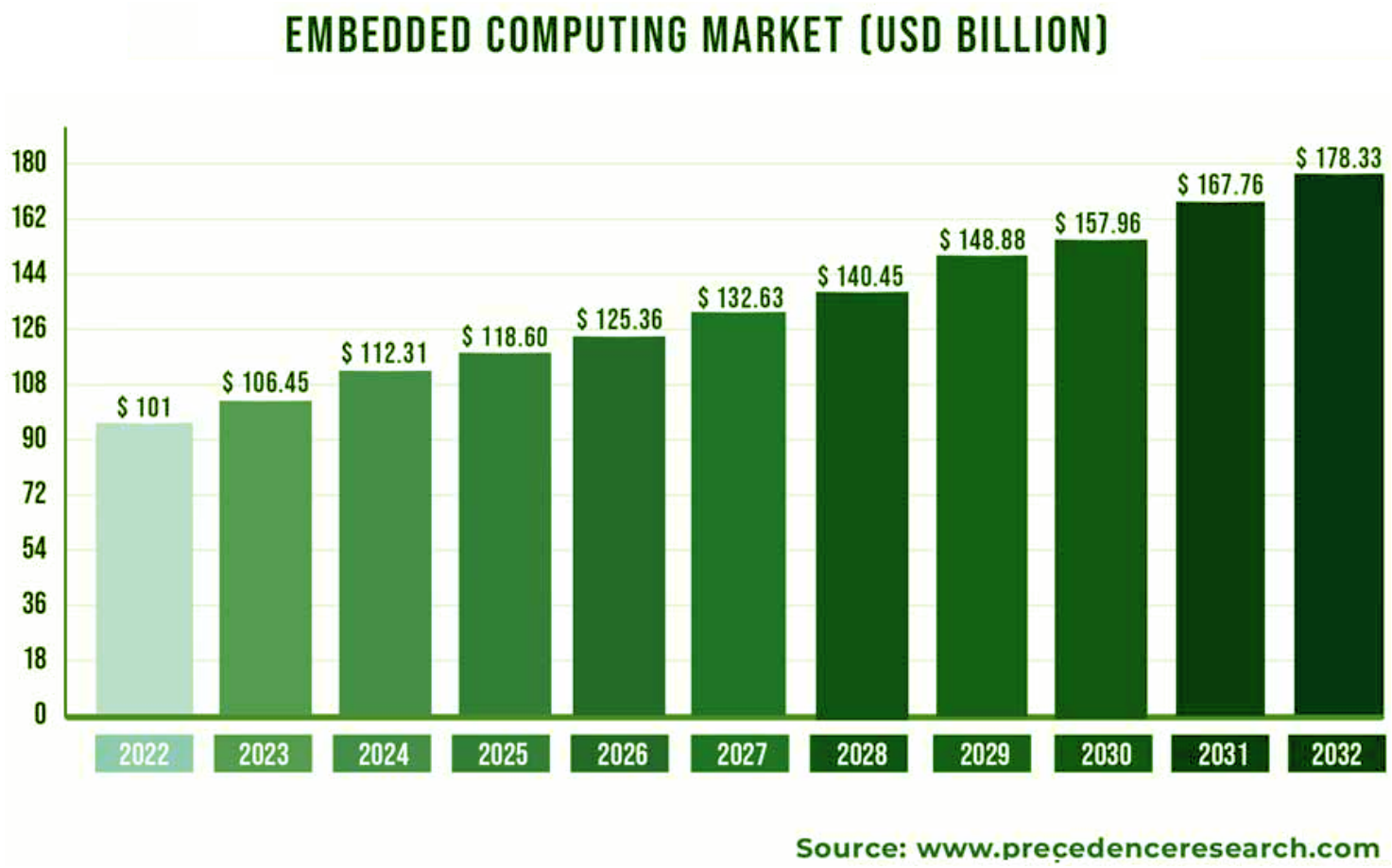
Evolution

- In the early days
- Today



Future

- Why are we interested in embedded systems?



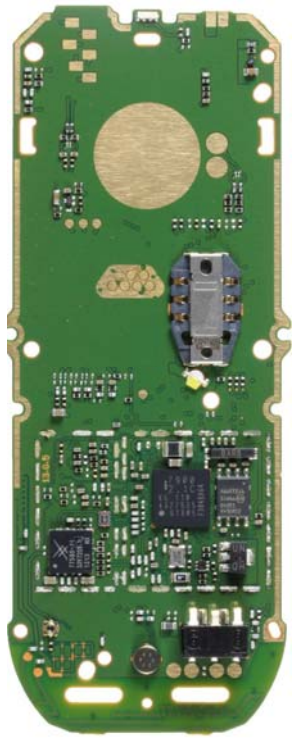
Shift of Industry

- What's the secret behind exponential revenue growth?



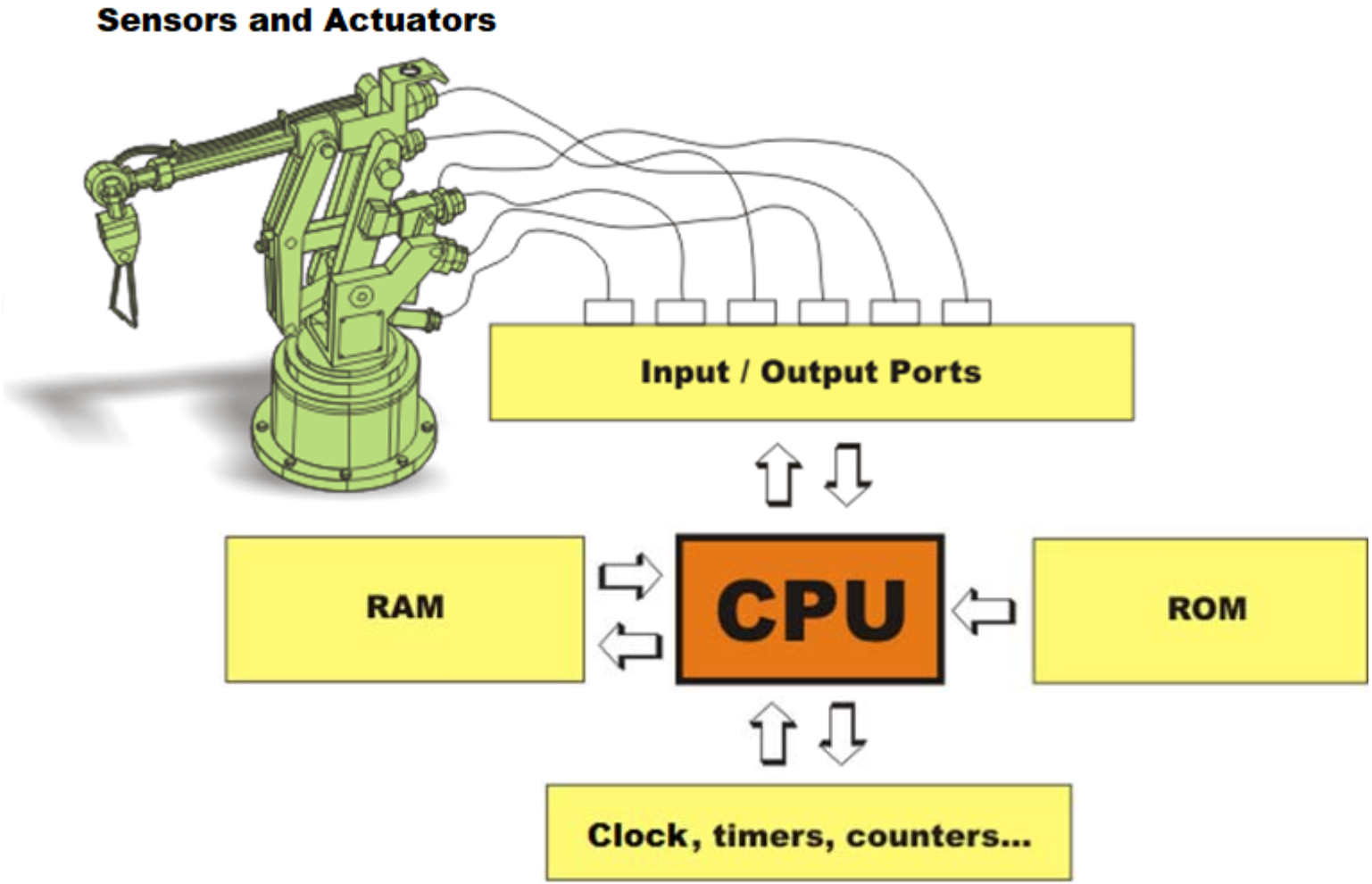






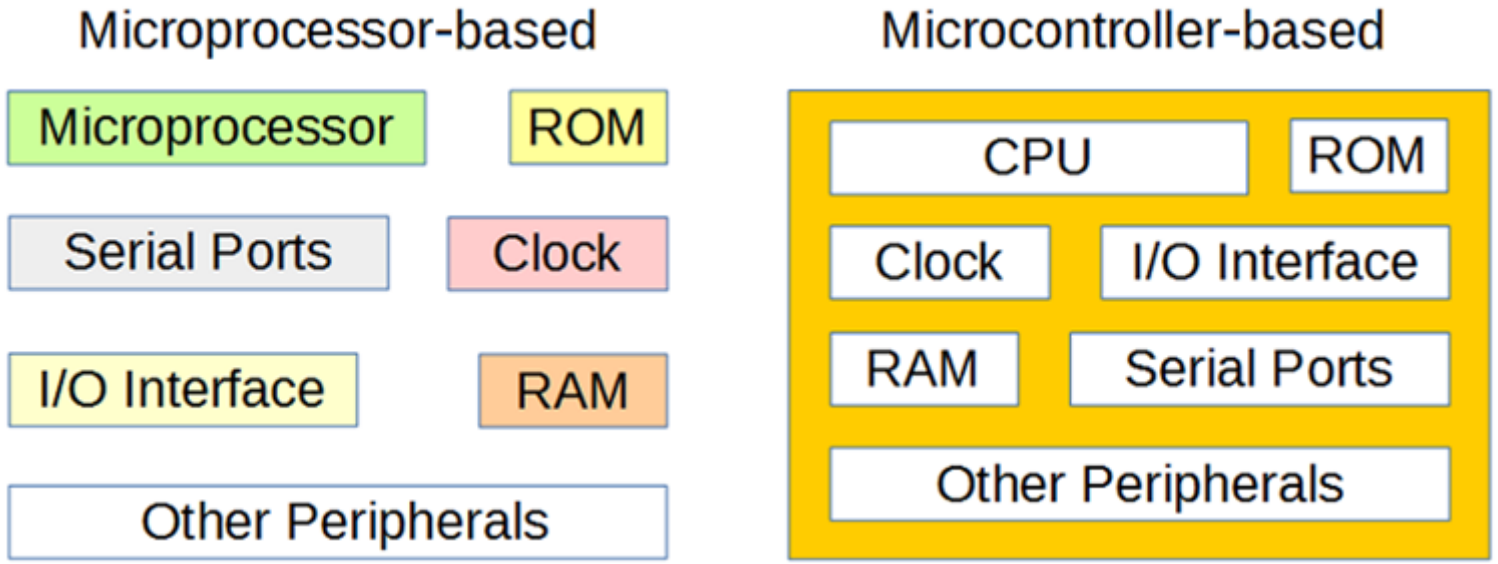
- What are the advantages?

Components


Sensors and Actuators
Input / Output Ports
RAM
CPU
ROM
Clock, timers, counters...

Implementations

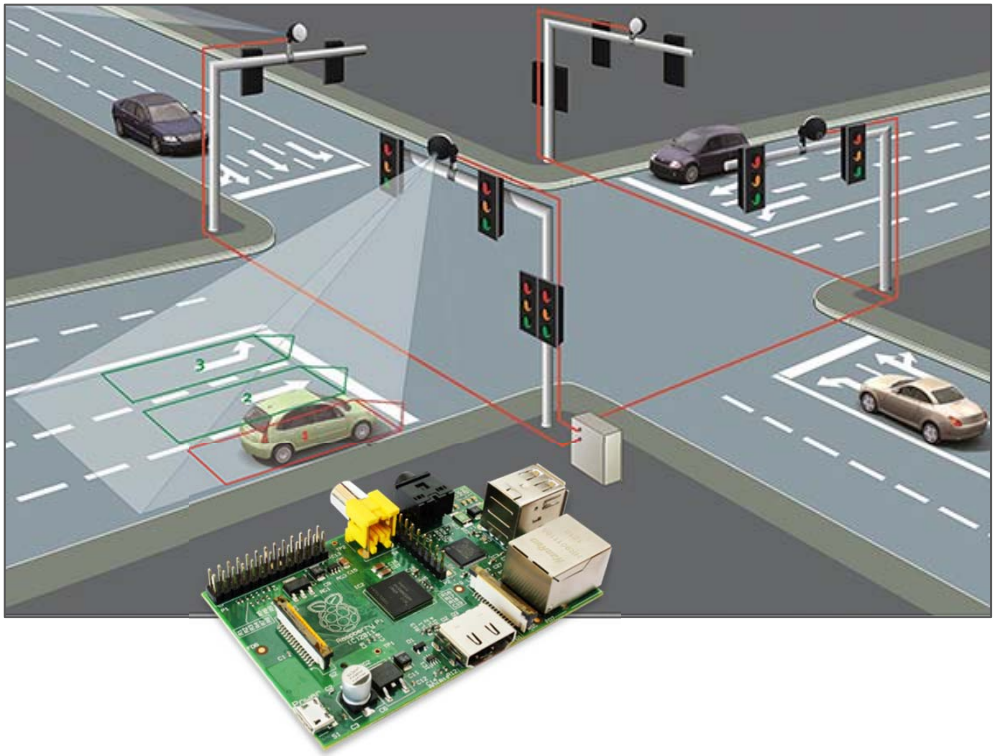
- Microprocessor-based vs. Microcontroller-based



  - What are the pros and cons?

Implementations

- Microprocessor-based or Microcontroller-based?

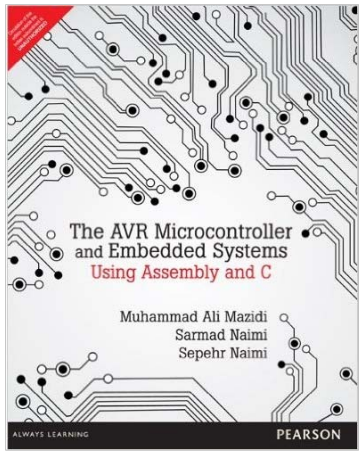


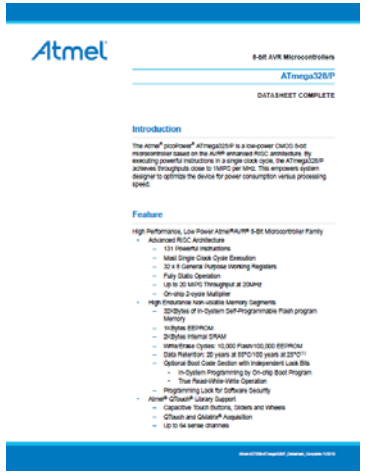


- Why are majority of embedded systems microcontroller-based?

Learning Materials

- Reference
  - *The AVR Microcontroller and Embedded Systems Using Assembly and C*, M. A. Mazidi et al., Pearson, 2017
  - *Microchip (Atmel) ATmega328P Datasheet*, https://www.microchip.com/en-us/product/ATmega328P



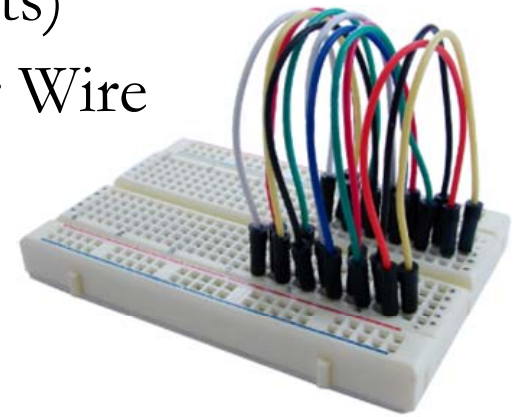


Learning Materials

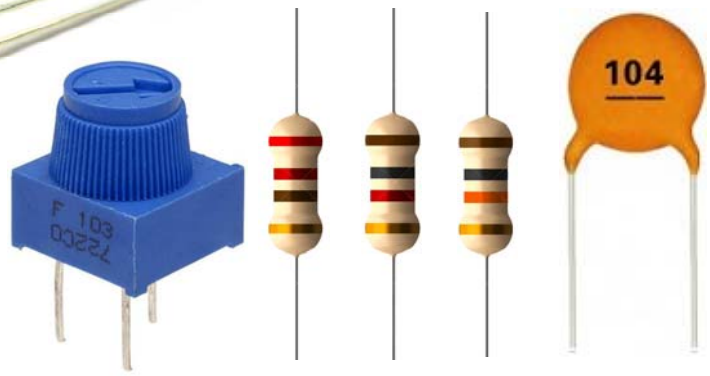


- Hardware
  - 1 × Arduino Uno Revision 3
  - 1 × Solderless Breadboard (400 contacts)
  - 20 × Male to Male Breadboard Jumper Wire
  - 2 × Micro Switch
  - 1 × 10kΩ Potentiometer
  - 10 × LED
  - 10 × 220Ω Resistor
  - 2 × 1kΩ Resistor
  - 2 × 10kΩ Resistor
  - 2 × 100nF Capacitor



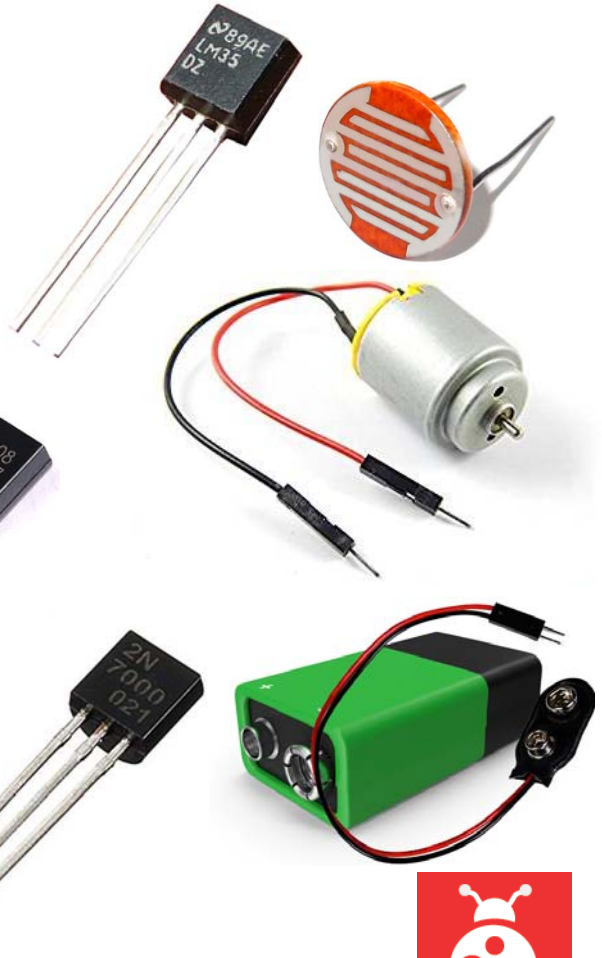






Learning Materials

- Hardware
  - 1 × Light Dependent Resistor (LDR)
  - 1 × LM35 Temperature Sensor
  - 1 × 6-9V DC Motor
  - 2 × 1N4001 Rectifier Diode
  - 2 × BC547 Transistor
  - 1 × 2N7000 Transistor
  - 1 × 9V Battery & Snap Connector
- Software
  - Microchip Studio from https://www.microchip.com
  - Arduino IDE from https://www.arduino.cc





Questions